

Linear Algebra Mini-Lesson

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1. Vectors and linear combinations

A vector can represent a quantity with magnitude and direction, or an ordered list of numbers. The two-dimensional vector $v = (2, 1)$ can represent a point or a displacement in the plane.

The expression $a u + b v$ is a linear combination of u and v .

Let $u = (1, 0)$ and $v = (0, 1)$. Then $3u + 2v = (3, 2)$. By changing the coefficients, these two vectors can describe different points in the plane.

Search verification keyword: cobalt-vector-17. Tests can verify that this unique phrase appears on page 1.

2. Matrix-vector multiplication

A matrix-vector product can be calculated by taking the dot product of each matrix row with the vector. It can also be interpreted as a linear combination of the matrix columns.

Matrix A	Vector x	Result Ax
[[1, 2], [3, 4]]	[5, 6]	[17, 39]

Using the columns, $Ax = 5(1, 3) + 6(2, 4) = (17, 39)$. This view shows that matrix-vector multiplication is a linear combination.

Search verification keyword: amber-matrix-42. A page-aware search result should cite page 2.

3. Check your understanding

Exercise 1: Given $u = (2, 1)$ and $v = (-1, 3)$, find $2u + v$.

Exercise 2: Given $A = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$ and $x = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$, find Ax .

Answers

Exercise 1: $2u + v = 2(2, 1) + (-1, 3) = (3, 5)$.

Exercise 2: $Ax = \begin{bmatrix} 2 \times 4 + 0 \times 2 \\ 1 \times 4 + 3 \times 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 10 \end{bmatrix}$.

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